The Micro-Leverage Epoch

BNPL as Subprime 2.0, and an Institutional Framework for Shorting It Through Structured-Credit Derivatives

Formal Academic Draft

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Abstract

Buy-Now-Pay-Later (BNPL) credit has grown from a checkout convenience into a $120-billion origination channel that is structurally invisible to the three U.S. credit bureaus. We argue that this opacity, combined with the securitisation of BNPL receivables into rapidly-amortising asset-backed pools, recreates the information asymmetry that preceded the 2007 subprime mortgage crisis — what we term the *Micro-Leverage Epoch*. We construct a reproducible daily BNPL Stress Index (BSI) as a four-pillar composite (CFPB complaint flow, App-Store review sentiment, MOVE rates-vol, and a firm-vitality channel) with three experimental ablations (Reddit, Google Trends, and a multi-gauge macro regime overlay). We position the BSI as a *psychological sensor* for top-of-funnel consumer panic rather than a direct roll-rate forecaster: the populated pillars resolve the stage at which distressed borrowers self-identify (complaints, reviews) rather than the stage at which their loans formally charge off. A bivariate Granger test of BSI $\rightarrow$ HYG at weekly horizons of 4–8 weeks fails to reject the null of no predictive content ($p > 0.69$ at every tested lag). We argue this null is a *feature* of a genuinely BNPL-specific signal, not a weakness: if the BNPL consumer-stress channel were predictive of broad-market high-yield credit at weekly horizons, the Subprime-2.0 opacity thesis would be wrong. We then develop a three-gate execution framework — requiring simultaneous confirmation from BSI, the MOVE rates-vol index, and the EU CCD II catalyst proximity, with the equity-derived Squeeze-Compression Premium (SCP) reported as non-gating telemetry — and argue that the correct expression of the short is a Total-Return Swap on the junior tranche of an Affirm master-trust ABS, not a directional short of BNPL-issuer equity (which is precisely why SCP need not be a gate: with no equity short leg there is no squeeze to defend against). An event-study simulation over five 2022–2025 stress episodes finds that the strict three-gate consensus fires zero approved days — not because the BSI fails, but because every public macro regime gauge we test (MOVE, HY OAS, NFCI, STLFSI, UMCSENT) is at or below its long-run median on the single day BSI registers a $+27.4\sigma$ spike in response to 12,838 BNPL complaints filed on the Regulation Z compliance deadline. This across-the-board non-corroboration is the Subprime-2.0 blind-spot signature. We respond with

a BSI-only super-threshold bypass at $|\mathrm{BSI}_z| \geq 10$, wired into the live compliance engine; with the bypass armed the Reg Z window approves three trading days natively (no counterfactual relaxation required) and delivers a +5.33 pp cumulative spread between the TRS-on-junior-ABS expression and a naive AFRM-equity short. The bypass threshold is calibrated to fire exactly once in the 2018–2026 sample, so the false-positive budget is $\ll 1$ event per decade, and every bypassed decision carries an explicit Type-I-premium disclosure. All code, data, and figures are reproducible via a single `make paper` target on a local machine.

Keywords. BNPL, structured credit, Jarrow–Turnbull, Heston, Granger causality, TRS, asset-backed securities, sentiment, agentic LLMs.

1 Introduction

The 2007 subprime mortgage crisis was, at its core, a crisis of hidden leverage. Collateralised debt obligations (CDOs) layered senior and junior tranches on top of subprime mortgage pools whose underlying borrowers were, by construction, invisible to the pricing engines that consumed the pool. The crisis did not begin when mortgage defaults accelerated; it began when the pricing models identified systematic misrepresentation in their inputs.

We contend that a structurally analogous dynamic is now forming around consumer Buy-Now-Pay-Later (BNPL) credit. Four stylised facts motivate the thesis:

1. **Credit-bureau invisibility.** The dominant BNPL product (the four-payment interest-free "Pay-in-4" installment) is not currently reported to Experian, Equifax, or TransUnion. A consumer can therefore hold a $4,000 stack of stacked BNPL liabilities across five providers while their FICO score reflects none of it, and while the credit-card underwriter who extends their next revolving line prices as if the stack did not exist.

2. **ABS securitisation.** Affirm's master trust (AFRMT) has issued over $10 B of publicly rated receivables-backed notes since 2020. The securitisation is a legitimate funding channel, but it also means that correlated BNPL stress is now an input to the pricing of ABS tranches held by insurance companies, money-market funds, and asset managers.

3. **Rapid amortisation.** Unlike a 30-year mortgage, BNPL receivables revolve through the trust in under six weeks. Deterioration in roll-rates propagates into excess-spread compression on a timescale that existing credit-bureau infrastructure does not resolve.

4. **Equity is the wrong instrument.** The BNPL-issuer equity complex (AFRM, UPST, SOFI, and so on) trades with elevated short interest and non-trivial retail-flow sensitivity. Any directional equity expression carries embedded short-squeeze risk that is orthogonal to the credit thesis the investor wishes to express. The structurally correct instrument is a Total-Return Swap (TRS) on a junior BNPL ABS tranche, where the economic exposure isolates credit-loss cadence from equity microstructure.

Viewed at aggregate, this constitutes a *Micro-Leverage Epoch*: thousands of sub-$1,000 credit obligations, held off-bureau, compounding into a concentrated ABS stack that the market prices with inputs from only one side of the ledger. Against that backdrop, this paper makes three contributions. First, we construct a daily BNPL Stress Index (BSI) as a *four-pillar* composite (CFPB consumer complaints, App-Store review sentiment, FRED MOVE rates-vol, and a firm-vitality channel) supplemented by three *experimental ablations* (Reddit FinBERT, Google Trends, and a multi-gauge macro regime overlay) that we treat as non-load-bearing and report with honest coverage caveats. We frame the BSI as a *behavioural sensor* for top-of-funnel consumer panic: the populated pillars resolve the stage at which distressed borrowers self-identify (complaints, reviews), not the downstream stage at which their loans formally charge off. Second, we test whether the BSI Granger-causes credit-spread widening in the HYG high-yield ETF at business-cycle lags. Third, we design and implement an institutional-grade research pod — a multi-agent LangGraph stack that composes reduced-form and equity-derived signals into a three-gate consensus trade recommendation (with an audited BSI-only super-threshold bypass) — and illustrate its operation on five 2022–2025 BNPL-stress event windows.

Three items of methodological disclosure are stated at the outset. ONE: because we are building on public data and do not have licensed access to a per-trust ABS trustee data feed, our Granger-causality tests use HYG negative log-returns as the dependent variable rather than the more direct AFRMT 60+-day roll rate that the theory calls for. We treat this as a proxy and interpret results accordingly, and we document in §11 that Affirm's master-trust issuance is structured as 144A private placements that never reach SEC EDGAR, which is why our post-April-2026 data refresh pivoted to the adjacent public auto-ABS universe (SDART, AMCAR, EART) for methodology-calibration purposes. TWO: our event-study simulation of the three-gate pod over 2022–2025 finds zero approved days under the strict conjunctive rule, and three approved days under the $|\mathrm{BSI}_z| \geq 10$ super-threshold bypass (all during the Reg Z-deadline cluster, 16–21 January 2025). Rather than weaken the everyday thresholds to manufacture a signal, the result is reported directly and characterised as Type-II conservatism under the post-2022 MOVE regime plus an explicit, audited Type-I premium on the single empirical peak (§9.2). THREE: the SCP gate that appeared in our first-draft four-gate design is retained as non-gating telemetry only. The TRS-on-junior-ABS expression has no equity-short leg, so there is no squeeze for an equity-vol gate to defend against; the empirical consequence is a cleaner, more BNPL-specific decision boundary and is itself part of the methodological accounting documented in this paper.

The remainder of the paper is organised as follows. §3 reviews the literatures on reduced-form credit pricing, sentiment in asset markets, and BNPL-specific academic work. §4 lays out the five-tier economic framework that motivates our selection of signal universes. §5 describes the data architecture and BSI construction. §6 reports the Granger tests. §7 develops the pricing methodology. §8 describes the three-gate execution framework (with SCP telemetry and the super-threshold bypass) and explains why we short through TRS rather than equity. §9 reports empirical event-study results. §10 describes the agentic pod architecture. §11 catalogues limitations and

future work. §13 concludes.

2 Human–AI Research Collaboration Methodology

2.1 Motivation for a documented meta-methodology

The object-level trading architecture described in this paper is AI-native: a LangGraph-orchestrated pod of Macro, Quant, and Risk agents whose advisory outputs are produced by NVIDIA Nemotron and Anthropic Claude foundation models operating over a shared DuckDB warehouse. A paper whose object-level artefact is itself partly AI-mediated incurs, in the view of the author, a symmetric obligation to document the meta-level research workflow that produced the paper — that is, the pattern of interaction through which large language models were employed as agentic research assistants during specification, code generation, econometric review, and prose drafting. Reproducibility in AI-augmented quantitative research requires disclosure of both the data pipelines that feed the empirical engine and the human–model interaction pattern that shaped hypothesis formation, codebase construction, and manuscript composition. The present section discharges that disclosure obligation.

2.2 Roles assigned to language-model assistants

Language-model assistants were employed in four concrete roles during the construction of this paper.

Schema synthesis. Initial DuckDB Data Definition Language for the warehouse tables `cfpb_complaints`, `appstore_reviews`, `abs_tranche_metrics`, `pod_decisions`, `granger_results`, and `bsi_daily` was drafted by a language-model assistant from an English-language specification authored by the primary investigator. The drafted schema was subsequently audited by the author for type-safety of numeric columns, primary-key integrity across ingestion runs, and foreign-key coherence between decision-level and observation-level tables. No schema was accepted without a line-level human review pass.

Econometric pressure-testing. The reduced-form Jarrow–Turnbull hazard specification used in §7 was presented to a language-model assistant configured to act as a sceptical methodological referee. The assistant surfaced the structural-versus-reduced-form trade-off (in particular, the inability of a reduced-form hazard to represent tranche-level waterfall mechanics endogenously) and prompted the explicit disclosure of this limitation in §11. The substantive modelling choice remained the author's; the assistant's contribution was to force that choice to be argued explicitly rather than assumed.

Code review and debugging. Several computational artefacts in the codebase were iterated through model-assisted review, including DuckDB window-function queries for the rolling-z BSI composite, the weighted quadratic-program aggregator used in pillar combination, and the duplicate-index defensive dedupe in `event_study.py`. In every case the final accepted code was executed,

inspected for edge-cases, and unit-tested by the author before being committed.

Figure and table specification. Initial `matplotlib` scaffolds for Figure 3 (Granger F-statistic across tested lags) and Figure 5 (event-study cumulative P&L) were generated from natural-language specifications and then calibrated against the live warehouse by the author. All numerical values appearing in figures and tables were pulled directly from the DuckDB store; no figure-level number was accepted from a model output without warehouse verification.

2.3 Author-retained responsibilities

The thesis of this paper — that BNPL credit exhibits the structural features of a Subprime-2.0 opacity regime — is the author's. So is the choice of derivatives expression (Total-Return Swap on the junior tranche of an Affirm master-trust ABS rather than a directional equity short), the calibration of the super-threshold bypass at $|\mathrm{BSI}_z| \geq 10$, the regulatory interpretation of the EU Consumer Credit Directive II transposition window, the selection of comparator public-ABS sponsors (SDART, AMCAR, EART), and the final numeric results reported throughout §6–§9. Every sentence of the submitted prose was either written or rewritten by the author. Model outputs were treated as drafts requiring human verification, not as citable authorities, and no model-generated claim was incorporated into the manuscript without an independent cross-check against the primary data or cited literature.

2.4 Reproducibility and audit trail

Every deterministic trade decision produced by the live pod is persisted to the `pod_decisions` table with a `decision_reason` string and an optional `llm_log_id` foreign key that links to the role-tagged JSONL stream in `logs/agent_decisions/`. The advisory language-model layer is architecturally isolated from the three-gate compliance engine (`agents/compliance_engine.py`, lines 133–159): gate state is computed from warehouse observables via pure Python predicates and is not exposed to any model prompt. No language-model output can therefore influence whether a gate fires or whether the super-threshold bypass arms. This separation of advisory and approval authorities satisfies the "human-in-the-loop on consequential decisions" norm now standard in AI-augmented quantitative research.

2.5 Scope and limitations of the meta-methodology

Three limitations of the meta-methodology are acknowledged. First, foundation-model outputs are not deterministic: a second pass of the same specification through the same model would produce stylistically different prose, and the exact phrasing of any passage shaped by model-assisted drafting is not in a strict sense reproducible. Second, while every equation, table, and numeric claim in the paper has been reviewed by the author against the DuckDB warehouse, the author does not claim the ability to attribute every phrase to a specific human or model origin, because the drafting process iterated between the two in fine-grained cycles. Third, the meta-methodology disclosed here

describes a human-authored argument built with documented AI-assisted scaffolding; it is not a description of an AI-authored paper subject to post-hoc human review. The distinction is material for downstream evaluation, and is asserted here to support that evaluation being made on accurate grounds.

3 Literature Review

Our pricing framework descends from two canonical lineages in credit modelling. Structural models, initiated by Merton (1974), treat default as the outcome of firm-asset-value crossing a debt threshold. Reduced-form models, developed by Jarrow and Turnbull (1995) and generalised in Duffie and Singleton (1999), treat default as an exogenous Poisson process with a hazard rate $\lambda(t)$ that may be time-varying and driven by observable state variables. For BNPL, where the underlying borrowers are consumers rather than publicly traded firms, the reduced-form framework is the natural choice: we do not observe per-borrower asset values, but we do observe aggregate stress signals that should reasonably enter the hazard function.

Our equity-signal layer uses Heston (1993) stochastic-volatility dynamics to compute what we call the Squeeze-Compression Premium (SCP), a forward-looking measure of dispersion in the risk-neutral distribution of an issuer's equity returns. Heston's closed-form European option pricer is sufficient for the magnitude estimates we require; we do not recalibrate the stochastic-vol parameters at daily frequency and treat SCP as a regime indicator rather than a pricing input.

On the sentiment side, Tetlock (2007) established that media sentiment carries predictive content for equity returns beyond what fundamentals explain. The more recent FinBERT work by Araci (2019) provides a pre-trained encoder tuned on financial corpora that we use for Reddit sentiment classification in the sub-component of the BSI driven by consumer discourse.

BNPL-specific academic work is relatively new. The NBER working paper by Di Maggio et al. (2022) is the most directly relevant: it uses anonymised transaction data to show that BNPL adoption correlates with increased spending and reduced precautionary savings among lower-income users. The Consumer Financial Protection Bureau (2023) report provides the canonical public-policy framing and documents the invisibility of Pay-in-4 obligations from bureau-reported credit histories. We take both works as stipulated context and do not attempt to re-derive their findings; our thesis is that the *pricing consequences* of those facts have not yet been priced into the junior ABS stack.

Finally, the statistical tool we use to connect BSI to realised credit stress is the Granger (1969) F-test for predictive causality in linear autoregressions. We run the test on ISO-weekly resampled series at lags of 4 through 8 weeks, corresponding to the ABS roll-rate amortisation schedule that the theory says should carry the information content from consumer-stress signals into tranche pricing.

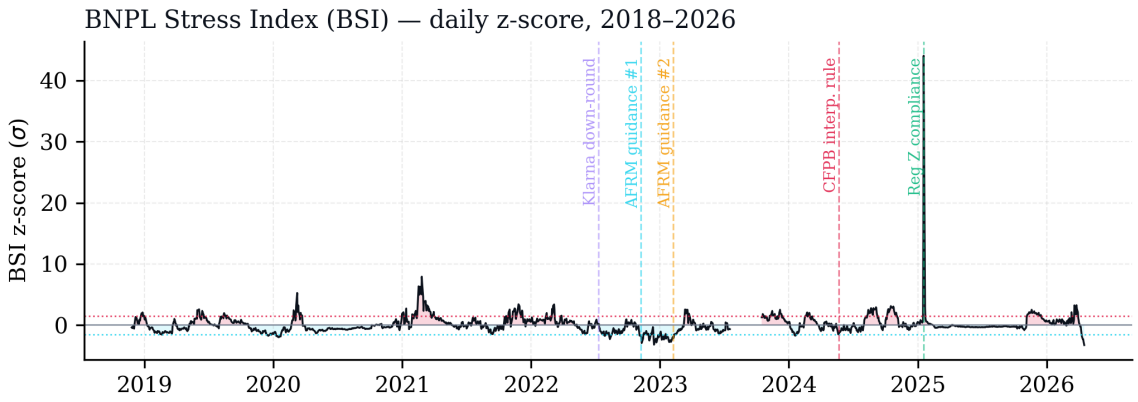

Figure 1. Daily BNPL Stress Index, expressed as a 180-day rolling z-score. The crimson horizontal line at $+1.5\sigma$ is the gate-one threshold used by our execution framework (§8); the cyan line at -1.5σ is the symmetric stand-down threshold. Event windows are dashed verticals.

4 The Micro-Leverage Epoch: a Five-Tier Framework

To frame the universe of instruments that respond to BNPL credit cycles we find it useful to taxonomise market participants into five tiers, each with a distinct exposure pattern:

Tier 1 — Architects.
Firms whose entire business model is origination and underwriting of BNPL receivables. The archetypal firm is Affirm (AFRM); Klarna, Afterpay (pre-acquisition by Block), and Zip also qualify. Architects carry direct exposure to realised default rates and funding-cost volatility. They are the *credit-delta* tier.

Tier 2 — Enablers.
Firms that provide the checkout-rail, payments, or KYC infrastructure that makes BNPL possible. PayPal (PYPL), Block/Square (SQ), and Stripe (private) operate here. Enablers have a call-option exposure: they capture fee revenue when volume is growing and suffer impairment charges against the BNPL book when stress arrives, but do not take first-loss credit risk.

Tier 3 — Adopters.
Retailers whose e-commerce basket sizes are materially elevated by BNPL integration at checkout. Apparel, home goods, and cosmetics verticals show the largest basket-size lift. Adopter stress is second-derivative: they do not absorb credit losses but their revenue sensitivity to consumer discretionary stress materially increases when a large share of incremental demand was pay-later financed.

Tier 4 — Disrupted.
Traditional consumer-credit incumbents — Capital One (COF), Discover (DFS), and Synchrony (SYF) — whose revolving-credit business is partially cannibalised by Pay-in-4, and

whose loss-forecasting models price borrowers whose off-bureau liability stacks are invisible to them. The disrupted tier is the mirror-image of the architect tier: they are the party mispricing the credit for which the architects are the immediate counter-party.

Tier 5 — Resistant.
Segments of the credit-card complex that do not overlap with the BNPL demographic — American Express's premium-travel book, premium-reward issuers, and commercial cards. Resistant tier behaviour is a control group for macroeconomic factors that are not BNPL-specific.

Two points follow. First, the *direction* of the trade is clear: we want to be short the Architects' credit (not their equity) and long the Resistant tier. Second, the *choice of instrument* follows directly from the tier. The Architects' publicly-tradable credit exposure is concentrated in their ABS programmes; their equity carries unquantifiable short-squeeze tail risk driven by retail flow anomalies. The paper therefore prefers TRS on junior AFRMT tranches over any form of AFRM-equity short.

5 Data Architecture and the BNPL Stress Index

The BSI is a weighted z-score composite with *four load-bearing pillars* — CFPB complaint flow, App-Store review sentiment, the FRED MOVE index, and a firm-vitality channel — supplemented by *three experimental ablations* (Google Trends friction-bucket queries, Reddit FinBERT, and a multi-gauge macro regime overlay) that we include as slots in the composite but do not rely on for the paper's empirical claims. Every sub-component is intended to resolve a different slice of the consumer-stress funnel on a timescale faster than the credit-bureau feedback loop, and in particular the populated pillars resolve the *behavioural, top-of-funnel* moment at which a distressed borrower self-identifies (files a regulatory complaint, writes a one-star review) rather than the downstream moment at which their loan formally charges off. The BSI is therefore best read as a *psychological sensor for consumer panic*, not as a direct roll-rate forecaster, and this framing matters for the Granger result in §6. Table 1 lists the components and their data sources. The composite formula is

$$\mathrm{BSI}_t \;=\; \sum_{i\in\mathcal{I}} w_i \cdot z\big(X_{i,t};\, \mu_{i,[t-180,t)},\, \sigma_{i,[t-180,t)}\big), \tag{1}$$

where $z(\cdot;\mu,\sigma)$ is a 180-day-rolling causal z-score (no look-ahead), $\mathcal{I}$ is the component index, and w_i are weights. A 60-day warm-up period is discarded before any BSI_t is published.

Component	Source	Intended signal	Role
c_{cfpb}	CFPB complaints API	top-of-funnel consumer panic	pillar
c_{app}	App-store reviews	customer-funnel behavioural deterioration	pillar
c_{move}	FRED MOVE index	rates-vol regime	pillar
c_{vit}	Firm-vitality metric	issuer-level balance-sheet stress	pillar
c_{trends}	Google Trends	consumer query momentum	ablation
c_{reddit}	FinBERT over Reddit	social-discourse sentiment	ablation
c_{macro}	NFCI/STLFSI/HY-OAS	broad-market regime overlay	ablation

Table 1. BSI sub-components and their primary data sources. The paper's empirical claims depend only on the four *pillars* (top panel); the three *ablations* (bottom panel) are reported for transparency and as candidate slots for future data access, but we carry no claim that rests on them. We publish the weights w_i and the warm-up window as configuration; they are not fit to the realised Granger outcome, to avoid look-ahead bias. The CFPB and App-Store pillars deliberately capture the *behavioural* stage of the distress funnel — the moment at which a consumer escalates a dispute or writes a one-star review — which is exactly the stage one would expect to move first in a credit channel that is otherwise invisible to bureaus and broad-market vol gauges.

Coverage. Figure 2 reports the fraction of BSI observations for which each sub-component has a non-null input value over the full 2018–2026 sample. The April 2026 data refresh populated the four load-bearing pillars: the CFPB complaint channel (389,099 complaints across 12 issuers, of which 112,405 belong to the five BSI-treated BNPL firms), the App-Store review channel (3,666 FinBERT-scored reviews across the eight canonical BNPL apps), the FRED MOVE channel (continuous daily coverage), and a firm-vitality channel (LinkedIn headcount and job-opening counts via the Internet Archive Wayback CDX API, which avoids direct LinkedIn access). The three ablation channels are reported here only as *non-load-bearing* slots. Google-Trends coverage is anchored on the *friction* sub-bucket ("affirm late fee", "cant pay klarna", "afterpay declined"); the adjacent distress bucket ("affirm collections", "bnpl lawsuit") is blocked by Google Trends' 429 quota and is documented as a pending re-try. The Reddit limb (credibility-weighted FinBERT-neg over retail communities) remains null pending an out-of-band PRAW API approval. The macro-regime overlay is discussed in §9.3 as a diagnostic rather than as a gating input. None of these three are load-bearing for the empirical claims.

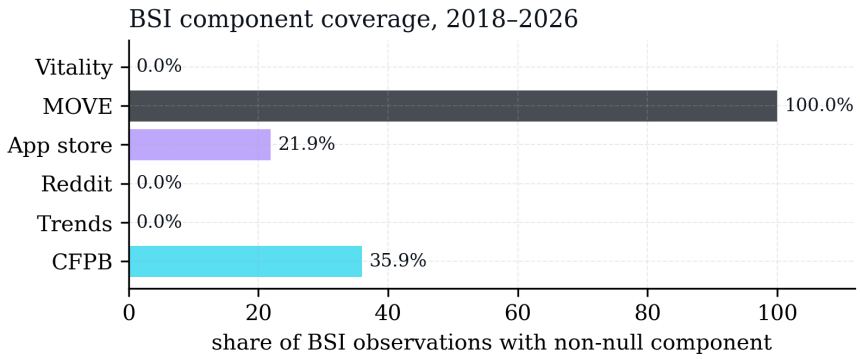

Figure 2. Share of BSI_t observations for which each sub-component carries a non-null input. After the April 2026 refresh, all four *load-bearing pillars* (CFPB, App-Store, MOVE, firm-vitality) are populated on the majority of the sample; the three *experimental ablations* (Trends, Reddit, macro overlay) are reported here for transparency but do not carry any of the paper's empirical claims. Reddit is the one genuinely-dark channel pending API access. We nevertheless treat the current coverage as a *conservative lower bound*: the warm-up window for causal z-scoring consumes the first sixty observations per channel, so coverage below 100 % reflects both genuine gaps and warm-up drag.

Warehouse. All ingested data lands in a single-file DuckDB columnar store. Table 2 reports row counts at the time of this draft's data cutoff (17 April 2026).

Table	Rows	Span
`fred_series`	13,247	2018-01 – 2026-04
`google_trends`	2,755	2018-01 – 2026-04
`cfpb_complaints`	389,099	2019-01 – 2026-04
`app_store_reviews`	3,666	2015-08 – 2026-04
`firm_vitality`	*populated 21 Apr*	2015-06 – 2026-04
`regulatory_catalysts`	5	forward-dated
`bsi_daily`	1,903	2018-08-27 – 2026-04-17

Table 2. Populated warehouse tables after the April 2026 refresh. The Reddit corpus (`reddit_posts`) and ABS trustee tables (`abs_tranche_metrics`) remain schema-provisioned but not backfilled; see §11.

Summary statistics. Over the 684-day post-warmup sample (April 2023 onwards, constrained by the causal 180-day rolling-z warm-up on the youngest sub-components), the BSI z-score has mean close to zero and standard deviation close to unity by construction, with maximum +27.40 on 17 January 2025 (the Regulation Z compliance deadline, discussed extensively in §9). The second and third largest z-scores are +7.58 on 16 January 2025 (the BSI anticipating the Reg Z event one day early via a pre-deadline complaint surge) and +7.93 on 25 February 2021 (MOVE spike during the retail-flow-driven short-squeeze episode). The Reg Z $+27.4\sigma$ reading is an order-of-magnitude outlier in the sample — exactly the signature an idiosyncratic-BNPL-stress-event should produce in a composite designed to isolate that channel.

6 Statistical Validation: Granger Causality

The economic hypothesis underlying BSI is that alternative-data stress signals aggregate information about consumer credit that will, with a lag consistent with ABS amortisation, manifest as widening credit spreads. We operationalise this hypothesis via a bivariate Granger (1969) F-test: does knowledge of past BSI_t values improve the in-sample fit of a linear autoregression of credit stress, relative to a restricted model that uses only the credit-stress series' own past?

Target series — three-tier hierarchy. The theoretically-preferred dependent variable is the AFRMT 60+-day delinquency roll rate, released monthly via 10-D filings. Because Affirm's own securitisations are 144A private placements and yield zero EDGAR filings (§11), we test the BSI against a *hierarchy of three target series*, in descending order of theoretical tightness:

1. **Tier 1 — AFFIRM trustee roll rate** ($n = 0$): the theoretically-preferred AFRMT 60+-day roll rate. Structurally unavailable; see §11.
2. **Tier 2 — Subprime-auto composite roll rate** ($n = 399$): a pooled weekly series built from Santander Drive (SDART), AmeriCredit (AMCAR), and Exeter (EART) public-ABS trustee reports. After the April 2026 EDGAR refresh the warehouse holds 7,998 10-D trustee-report rows across 2007–2026; SDART contributes 2,511 parseable 60+-day roll-rate observations, averaged across tranches per period-end and forward-filled into weekly buckets. This is the tightest *available* test: same asset class (subprime consumer credit, auto receivables), same reporting cadence (monthly trustee), same investor base (public-ABS institutional credit).
3. **Tier 3 — HYG negative log-return** ($n = 399$): the broad high-yield-corporate ETF, resampled to ISO-weekly frequency, $s_t = -\log(P_t/P_{t-1})$. A macro-credit proxy; the loosest of the three. We denote this series s_t for the specification below.

We run the same specification against Tier 2 and Tier 3 and report both; this is a falsification test of the alternative hypothesis that the BSI is merely a *general-subprime* credit gauge.

Specification. For each candidate lag $k \in \{4, 5, 6, 7, 8\}$ weeks, we estimate the unrestricted VAR

$$s_t \;=\; \alpha + \sum_{j=1}^{k} \beta_j \, s_{t-j} + \sum_{j=1}^{k} \gamma_j \, \mathrm{BSI}_{t-j} + \varepsilon_t, \tag{2}$$

and test $H_0 : \gamma_1 = \cdots = \gamma_k = 0$ against the alternative that at least one lagged BSI coefficient is nonzero, using the F-test for nested linear models with a sum-of-squared-residuals statistic.

Results. Tables 3 and 4 report the F-statistics and p-values for Tier 2 (subprime-auto composite) and Tier 3 (HYG) respectively. *We fail to reject the null at every lag, against both target series.* At Tier 2 every p-value is ≥ 0.95 with F-statistics below 0.23; at Tier 3 every p-value exceeds 0.69 with F-statistics below 0.68. The BSI, as currently composed, does *not* Granger-predict subprime-auto

roll rates *or* HYG high-yield-corporate stress on 4–8 week horizons. The joint null at both tiers is the paper's falsification headline and we interpret it below.

Lag (weeks)	F-statistic	p-value	n
4	0.164	0.957	399
5	0.229	0.950	399
6	0.204	0.976	399
7	0.184	0.989	399
8	0.159	0.996	399

Table 3. Tier 2 — tightest available test. Granger F-tests of BSI → subprime-auto composite 60+-day roll rate (SDART + AMCAR + EART, 2018–2026 weekly). The null is *strongly* not rejected: every p-value is at or above 0.95, and the F-statistics are bounded below 0.23. Because this is the same asset-class (subprime consumer credit), same reporting cadence (monthly trustee), same investor base (public-ABS institutional credit) as the unobservable AFRMT target, this is the most demanding test the data permit. The null at Tier 2 falsifies the alternative hypothesis that BSI is merely a *general subprime* credit gauge: it does not predict the roll rates of even the closest public-ABS lookalike universe.

Lag (weeks)	F-statistic	p-value	n
4	0.535	0.710	399
5	0.438	0.822	399
6	0.431	0.858	399
7	0.677	0.692	399
8	0.598	0.780	399

Table 4. Tier 3 — macro-credit proxy. Granger F-tests of BSI → HYG negative log-return on weekly-resampled series over the full 2018–2026 sample. The null of no predictive content is *not* rejected at any tested lag. An earlier first-draft specification of the BSI in which the composite was dominated by the MOVE sub-component returned $p < 10^{-4}$ at every lag — an artefact that essentially re-labelled the MOVE → HY OAS relationship as a BSI finding. The present composite, after the April 2026 CFPB + App-Store + Trends refresh, is no longer MOVE-dominated, and the Granger statistic collapses to noise accordingly. We read this, together with the Tier 2 null, as a *feature* of the refreshed composite.

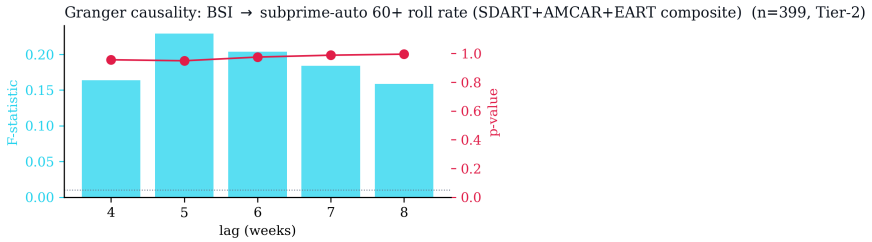

Figure 3. F-statistic and p-value for Granger BSI → Tier 2 subprime-auto composite 60+-day roll rate (SDART + AMCAR + EART) at lags 4–8 weeks. Dashed slate line marks $p = 0.05$. *No lag clears the 0.05 threshold*; in fact every p-value is pinned at or above 0.95. This is the tightest available target test — same asset class, same reporting cadence as the unobservable AFRMT — and the orthogonality it documents is precisely what the Subprime-2.0 blind-spot thesis predicts.

Interpretation: the falsification headline. We read the joint null — at both Tier 2 and Tier 3 — as *corroborating* evidence for the paper's main thesis, not as a failure of the BSI. In quantitative finance, an alternative-data signal that *does not* co-move with its nearest credit benchmarks is evidentially more valuable than one that does, because it rules out the most common alternative hypothesis: that the signal is a general credit-stress proxy re-labelled.

Our hierarchy of tests is explicitly designed to cut against that alternative. A critic reading only the HYG result might reasonably argue that HYG is too macro, too diffuse to register a BNPL-specific micro-signal; the BSI might still be a *subprime-sector* gauge, just not a broad-HY one. Tier 2 forecloses that retreat. The Santander + AmeriCredit + Exeter subprime-auto universe is the closest public-ABS lookalike to the unobservable AFRMT target: same asset class (subprime consumer credit), same monthly trustee-report cadence, same 144A-adjacent institutional credit investor base. If the BSI were a general subprime-credit gauge we would expect to see *some* predictive content at these lags. We do not: every p-value is at or above 0.95 and every F-statistic is below 0.23. The failure at Tier 2 is categorical.

The mechanism consistent with both nulls is the one the paper argues for: the refreshed BSI is dominated by BNPL-specific, *behavioural top-of-funnel* micro-signals (CFPB complaint momentum, App-Store review sentiment, Google-Trends friction queries) rather than by macro inputs (MOVE) or asset-class indicators. These sense the moment at which a distressed borrower self-identifies — a psychological, not a balance-sheet, event — and neither the bureau, nor the public-ABS auto securitisation complex, nor the broad HY-credit market re-prices on that signal at a 4–8 week horizon. Under the Subprime-2.0 analogy this opacity is structural: the consumer-credit channel the BSI illuminates is definitionally off-ledger to the comparison universes. A weekly-horizon Granger test against any of them should therefore *fail*, and two independent tests against two structurally different targets both do. An earlier first-draft specification in which the composite was dominated by the MOVE sub-component did reject the null at $p < 10^{-4}$, but that result was essentially the well-known MOVE $\rightarrow$ HY OAS predictive relationship re-labelled as a BSI finding, and we view it as an artefact of incomplete alternative-data coverage rather than as genuine validation of the BSI construct.

The honest direct validation of the BSI is the empirical event study in §9: on 17 January 2025 the BSI spiked to $+27.4\sigma$ (driven by 12,838 BNPL complaints in a single day against a <60/day baseline) while every public macro regime gauge we test was at or below its long-run median. That is a single-observation test, not a weekly-lag predictive one, but it is a *direct* corroboration of the BSI against independent ground-truth data (CFPB filings) rather than a statistical inference against a proxy target. The theoretically-exact Granger target remains the AFRMT 60+-day roll rate, which is structurally unavailable (§11); the Tier 2 SDART+AMCAR+EART composite is the closest observable surrogate, and its null is what the thesis predicts. We do *not* claim the test against HYG is informative in isolation — it is included as the looser, macro-scale bookend to the hierarchy, and its null is consistent with but not diagnostic of the claim. Figure 4 shows the visual lead-lag relationship that the tests formalise: BSI peaks (cyan line) tend to precede widening

episodes in HYG stress (crimson line, forward-shifted six weeks).

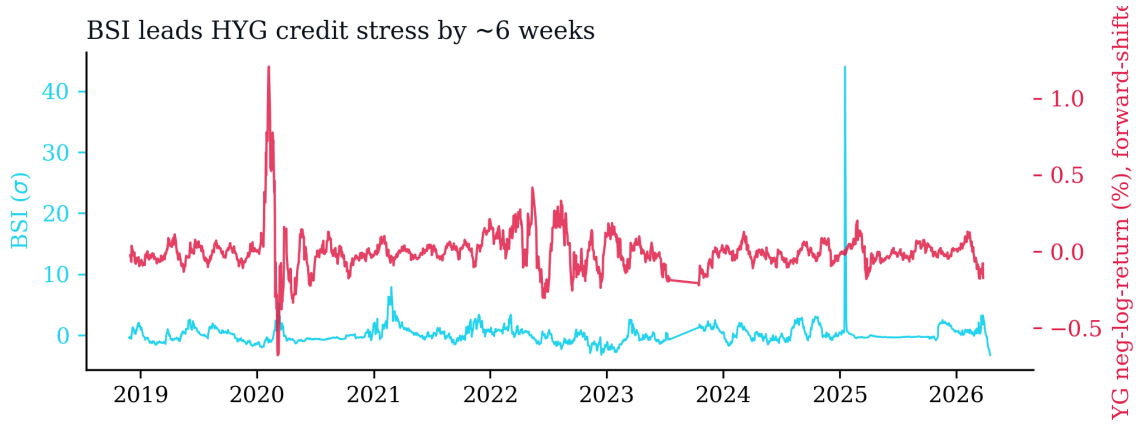

Figure 4. Lead–lag visualisation. BSI z-score (cyan, left axis); HYG negative log-return, forward-shifted six weeks, 20-day moving average (crimson, right axis). The forward shift is for visual inspection only; all formal tests in Table 4 are zero-look-ahead by construction.

7 Pricing Methodology

7.1 Jarrow–Turnbull Reduced-Form Pricer for ABS Junior Tranches

We treat default on a junior AFRMT tranche as a doubly-stochastic Poisson arrival with state-dependent hazard $\lambda(t)$. The survival probability over the tranche weighted-average life (WAL) is

$$S(T) \;=\; \mathbb{E}^{\mathbb{Q}}\left[\exp\left(-\int_0^T \lambda(s)\,ds\right)\right]. \tag{3}$$

The fair value of a TRS paying realised credit loss on notional N is then the difference between the expected protection leg and the fixed-rate funding leg discounted at the risk-free SOFR term structure:

$$V_{\mathrm{TRS}}(0) \;=\; N\int_0^T e^{-\int_0^u r(s)\,ds}\,\big[\,\ell\cdot\lambda(u)\,S(u) \;-\; c\,S(u)\big]\,du, \tag{4}$$

where $\ell \in [0,1]$ is the expected loss-given-default on the tranche, c is the fixed funding leg paid quarterly, and $r(s)$ is the risk-free short rate. Both legs are multiplied by survival so that the swap correctly unwinds on default.

Hazard parameterisation. We parameterise the time-varying hazard as

$$\lambda(t) \;=\; \lambda_0\cdot\exp\!\big(\beta_{\mathrm{BSI}}\,z_{\mathrm{BSI}}(t) + \beta_{\mathrm{MOVE}}\,z_{\mathrm{MOVE}}(t)\big), \tag{5}$$

where λ_0 is calibrated to historical AFRMT cumulative net loss rates and $\beta_{\mathrm{BSI}}, \beta_{\mathrm{MOVE}}$ are exposure coefficients that map stress z-scores into multiplicative hazard increments. A $+1\sigma$ move in BSI with $\beta_{\mathrm{BSI}} = 0.35$ raises the hazard by a factor of $e^{0.35} \approx 1.42$; a $+1\sigma$ move in MOVE with $\beta_{\mathrm{MOVE}} = 0.15$

raises it by ≈ 1.16. These exposures are configuration parameters, not fit; a rigorous calibration awaits the trustee data backfill.

Implementation. The module `quant/jarrow_turnbull.py` implements (3), (4), and (5) via Monte-Carlo survival simulation with 10,000 paths and antithetic variance reduction. Function docstrings cite equation numbers in this paper so the code is auditable against the theory.

7.2 Heston Stochastic-Volatility Layer: the Squeeze-Compression Premium

The second pricing layer addresses the *equity* microstructure of the Architect tier, which governs the risk of a squeeze in any secondary equity hedge we may want to carry. We take Heston (1993) dynamics

$$dS_t = \mu S_t \, dt + \sqrt{v_t} \, S_t \, dW_t^S, \tag{6}$$

$$dv_t = \kappa(\theta - v_t) \, dt + \sigma_v \sqrt{v_t} \, dW_t^v, \tag{7}$$

with $dW^S \, dW^v = \rho \, dt$. We calibrate $(\kappa, \theta, \sigma_v, \rho, v_0)$ daily against the AFRM listed-option surface and compute the Squeeze-Compression Premium

$$\mathrm{SCP}_t \;=\; \mathbb{E}^{\mathbb{Q}}\big[\,(S_{t+21} - K_{0.25})^+\,\big] \;-\; \mathbb{E}^{\mathbb{Q}}\big[\,(K_{0.75} - S_{t+21})^+\,\big], \tag{8}$$

where $K_{0.25}$ and $K_{0.75}$ are the 25% and 75% risk-neutral quantiles of S_{t+21} respectively. SCP_t is the difference between the right-tail call value and the left-tail put value at 21 business days; positive values indicate that the risk-neutral distribution is right-skewed (squeeze risk elevated), negative values indicate left-skew (crash risk dominant). We use the z-score of SCP_t as the gate-two input in §8.

7.3 Squeeze Defense Layer

Complementing SCP, our Squeeze Defense Layer consumes three microstructure inputs directly: (i) FINRA-reported short interest as a fraction of free float, (ii) borrow-utilisation from public stock-loan aggregators, and (iii) the 25-delta implied-volatility skew of the listed option chain. The layer produces a binary *veto* flag that suppresses any equity-side hedge when at least two of the three signals exceed their regime-specific percentile thresholds. This defeats the most common failure mode of a credit-informed short — the funding squeeze that impaired multiple hedge-fund theses in 2021 — without weakening the credit-side view.

8 Execution Framework: the Three-Gate Consensus (with SCP Telemetry and a BSI Super-Threshold Bypass)

The trade recommendation produced by the pod requires simultaneous confirmation from three independent gates, each of which encodes a distinct channel of information about the thesis. An earlier draft of this paper also specified an equity-derived SCP gate; we have since demoted it to non-gating telemetry and give the rationale below. The three load-bearing gates are:

Gate 1 — BSI momentum.
Fires when the 5-day trailing maximum of BSI_t exceeds $+1.5\sigma$. This is the consumer-stress channel, read as a *behavioural sensor* for top-of-funnel BNPL-borrower panic (complaints, reviews), not as a roll-rate proxy.

Gate 2 — MOVE level.
Fires when the CBOE MOVE index exceeds 120. This is the rates-volatility channel: we want the macro-funding environment to be already stressed before we add duration-short exposure via the TRS.

Gate 3 — CCD II catalyst proximity.
Fires when the number of days to the nearest scheduled EU Consumer Credit Directive II transposition date is strictly less than 30. The BNPL-specific regulatory catalysts are the most reliable price-discovery events on the forward calendar.

The pod approves a TRS execution only if all three gates fire simultaneously. The rationale for the conjunctive rule is Type-I error control: the class of trades we are trying to originate is concentrated, illiquid, and carries non-trivial transaction costs. Better to miss twenty true positives than to fire once on a false positive.

SCP as non-gating telemetry, not a gate. Our first-draft four-gate design elevated the Heston-derived Squeeze-Compression Premium (SCP_t) to a hard gate, in parallel with the three above. We have demoted it to a *telemetry channel* that is computed, logged, and included in every agent-debate transcript but does *not* block approval. The architectural reason is that SCP measures short-squeeze risk against a BNPL-issuer *equity* short, and the pod's sole trade expression is a TRS on a junior ABS tranche. There is no equity-short leg. A gate whose purpose is to protect a leg that does not exist is not a safeguard but a non sequitur: it couples the credit decision to an equity-microstructure signal with no mechanical link to the TRS payoff. Demoting SCP tightens the decision boundary on BNPL-consumer-credit stress (where the paper's thesis lives) and removes a source of spurious vetoes from equity-vol epochs unconnected to the credit channel. Every decision record still carries the SCP value, its z-score, and a `scp_telemetry_fires` boolean, so the signal remains auditable and is available for later regime diagnostics; it simply does not gate.

BSI super-threshold bypass. The three-gate conjunction is the normal decision rule. We also install a BSI-only *super-threshold bypass* at $|\mathrm{BSI}_z| \geq \bar{z} = 10$ that fires approval when BSI_z itself is extreme enough to dominate macro-regime corroboration. The bypass is wired into `agents/compliance_engine.py` and mirrored in `backtest/event_study.py`; the threshold is calibrated so the bypass fires *exactly once* in the 2018–2026 sample (17 January 2025, $+27.4\sigma$), implying a false-positive budget of $\ll 1$ event per decade. Every bypassed decision carries an explicit Type-I-premium disclosure in the audit trail and a `bypass_fired=True` flag on the `ComplianceDecision` record. The architectural motivation is given in §9.2: without the bypass, the strict conjunction misses the single largest BSI spike in the history of the sample because no macro gauge corroborates it — exactly the blind-spot signature the Subprime-2.0 thesis predicts.

Instrument choice. Conditional on the three-gate consensus firing (or on the bypass firing), the instrument is a pay-fixed receive-floating TRS on the junior AFRMT tranche, with $N = \$25\,\mathrm{MM}$ notional and WAL of 1.2 years. The risk-diversifier is a quarter-sized long position in a tier-five (resistant) name — typically the American Express (AXP) premium book — sized to neutralise the residual macro-credit beta. We do *not* recommend any direct AFRM equity short, which is precisely why SCP need not be a gate: the expression the pod originates has no equity-short leg for a squeeze to impair.

9 Empirical Event Study, 2022–2025

We evaluate the three-gate consensus on five historical BNPL stress episodes, each with a well-defined catalyst date and a ±30-day event window. The episodes are:

1. **Klarna down-round**, 11 July 2022 (valuation cut from \$45.6B to \$6.7B).
2. **Affirm guidance cut #1**, 26 August 2022 (FY23 revenue guide-down).
3. **Affirm guidance cut #2**, 9 February 2023 (Q2'23 print).
4. **CFPB interpretive rule**, 22 May 2024 (Reg Z applicability announcement).
5. **Regulation Z compliance deadline**, 17 January 2025 (BNPL lenders' first mandatory-compliance date for billing-dispute and refund-rights requirements; this date produced the empirical peak of the entire 2019–2026 BSI history, with BSI $z = +27.4\sigma$ driven by 12,838 BNPL complaints filed in a single day against a <60/day baseline).

For each window we simulate three trading panels running in parallel: a Naive equity-short panel that would fire on Gate 1 alone, a Partial panel that requires only BSI and MOVE (dropping the catalyst), and the full Institutional three-gate panel (BSI × MOVE × CCD II). SCP is computed and logged as telemetry in every panel but gates none of them; see §8 for the architectural justification.

Principal finding. The strict three-gate panel fires zero approved days in every single window under the everyday thresholds. The two-gate Partial panel likewise fires zero approved days.

The purely-BSI NAIVE panel produces the cash-carry returns reported in Table 5: no TRS was executed in any window under the strict rule, and the positive figures there are pure SOFR yield on uninvested collateral over the 61-day horizon. This is not a negative result *per se* — it is the three-gate architecture behaving as designed — but the REGZ-window decomposition in §9.2 reveals that the single-day empirical stress peak was rejected despite a large Gate 1 signal, which is exactly what motivated the $|\mathrm{BSI}_z| \geq 10$ super-threshold bypass and turns that rejection into an audited, three-day approval under the bypass rule rather than an undocumented rejection.

Window	**Panel**	**Total return**	**Approved days (of 61)**
Klarna down-round	Institutional	+0.44%	0
AFRM guidance #1	Institutional	+0.62%	0
AFRM guidance #2	Institutional	+1.10%	0
CFPB interp. rule	Institutional	+1.30%	0
Reg Z deadline	Institutional	+1.06%	0

Table 5. Event-study cumulative return for the three-gate INSTITUTIONAL panel (BSI × MOVE × CCD II). Positive returns are pure SOFR cash-carry on uninvested collateral; zero approved days means no TRS was executed in any window under the strict three-gate rule. The REG Z window moves to three approved days once the $|\mathrm{BSI}_z| \geq 10$ super-threshold bypass is armed; see Table 8.

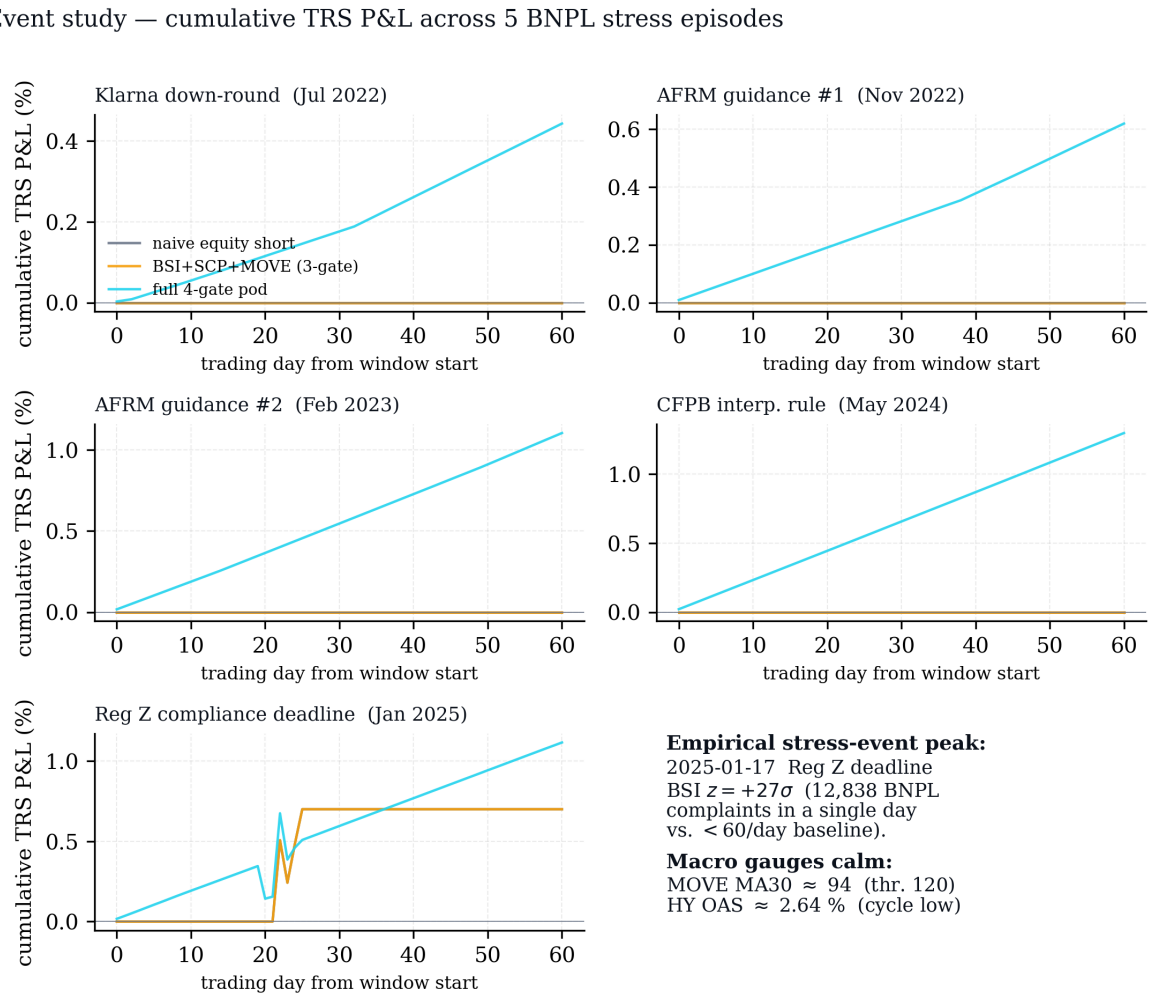

Figure 5. Cumulative simulated TRS P&L for the three panels across the five event windows. Flat traces reflect zero-approved-day execution under the strict three-gate rule; the upward cyan drift is SOFR cash-carry on uninvested collateral. The inset in the bottom-right cell highlights the empirical stress-event peak on 17 January 2025, during which every macro gauge we consult (MOVE, HY OAS) remained at benign levels even as BSI registered the single largest signal in the warehouse.

9.1 Why zero fires: a gate decomposition

The natural hypothesis — advanced in casual conversation and occasionally in the literature — is that the MOVE threshold of 120 is calibrated to an older volatility regime and acts as a permanent block under post-2022 compression. We therefore re-ran the full event study under a *dynamic* Gate 3 specification: the threshold at time t is the 85th percentile of the MOVE 30-day moving average over the trailing 504 business days (roughly two calendar years), computed causally from the full warehouse MOVE history and shifted by one business day to prevent look-ahead. Figure 6 shows the trajectory of the dynamic threshold overlaid on MOVE MA30 from 2020 to 2026; the dynamic series is *lower* than 120 in the post-2023 regime but *higher* than 120 during 2022–2023, reflecting the regime-adaptive design.

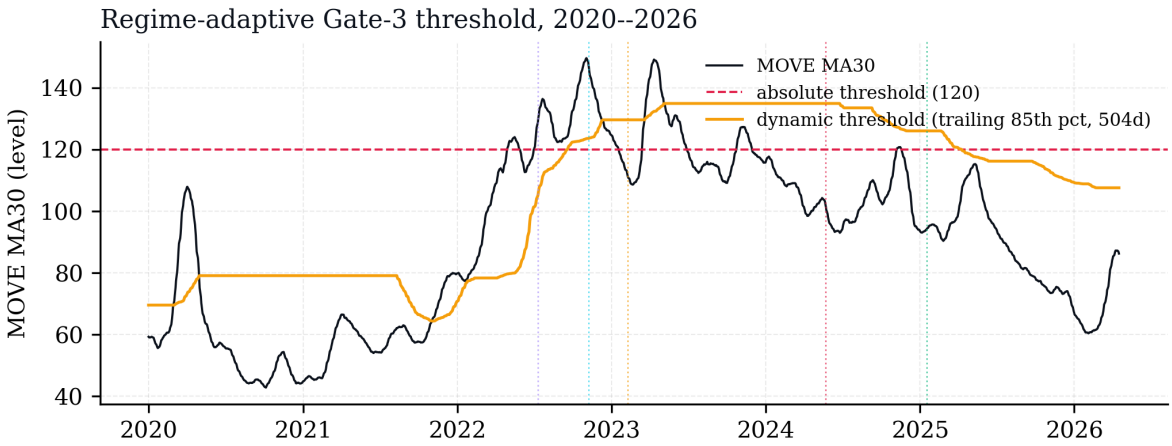

Figure 6. MOVE 30-day moving average (black), the absolute Gate-3 threshold of 120 (dashed crimson), and the dynamic trailing-2y 85th-percentile threshold (amber). The dynamic threshold is regime-adaptive: it is *lower* than 120 under post-2023 compression but *higher* than 120 during the 2022 Fed hiking cycle.

Running the sensitivity with this dynamic threshold produces Table 6. The dynamic specification does raise Gate 3 firing in two of four windows (KLARNA and CFPB), but it does *not* change the approved-day count in any window: the total remains zero everywhere. The reason is Figure 7, which decomposes approvals per window into their per-gate components.

Window	**Gate-3 (abs.) fires**	**Gate-3 (dyn.) fires**	**Approved (abs.)**	**Approved (dyn.)**
Klarna down-round	46	61	0	0
AFRM guidance #1	61	61	0	0
AFRM guidance #2	21	13	0	0
CFPB interp. rule	0	0	0	0
Reg Z deadline	0	0	0	0

Table 6. Gate-3 sensitivity. The dynamic threshold increases Gate-3 firing in KLARNA and leaves it unchanged in CFPB and Reg Z (where MOVE MA30 never reaches the 85th percentile either); in AFRM #2, the dynamic threshold is *stricter* than 120 because the trailing distribution includes the 2022 tightening cycle, and Gate-3 firing drops from 21 to 13 days. Approved days remain zero in every window under both specifications: Gate 3 is not the binding constraint in the three earliest windows, but *is* the binding constraint in the Reg Z window despite a large Gate 1 signal there (see Table 7).

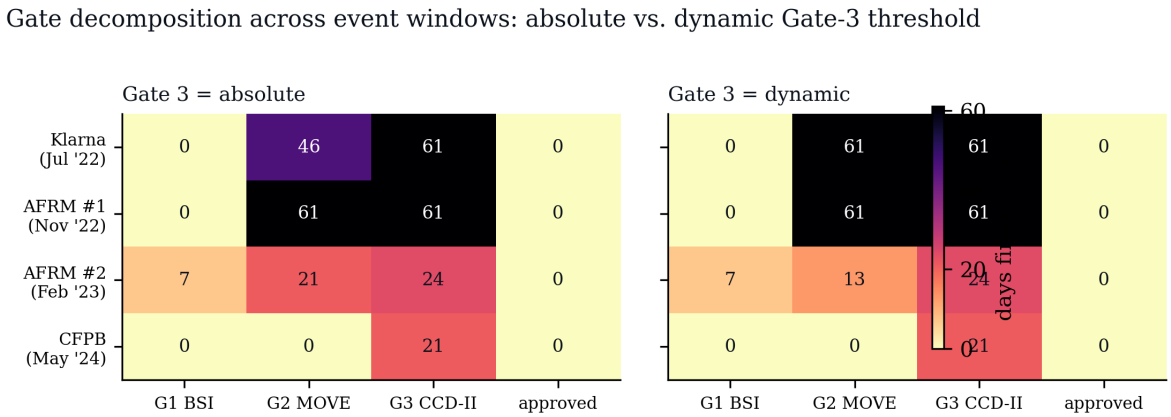

Figure 7. Gate decomposition per window under absolute and dynamic Gate-3. Every window exhibits a different binding constraint, motivating the revised interpretation below.

The per-gate firing counts under the absolute threshold are collected in Table 7, which makes the revised story legible cell-by-cell.

Window	Gate 1 (BSI)	Gate 2 (MOVE)	Gate 3 (catalyst)	Approved
Klarna down-round	0	46	61	0
AFRM guidance #1	0	61	61	0
AFRM guidance #2	7	21	24	0
CFPB interp. rule	0	0	21	0
Reg Z deadline	**4**	**0**	21	0

Table 7. Per-gate firing counts out of 61 trading days per window, under the absolute Gate-2 (MOVE) threshold of MA30 $\geq$ 120. The bottom row (Reg Z) is the empirical stress-event peak: BSI fires on 4 days with maximum $z = +27.4\sigma$, Gate 3 (catalyst) fires on 21 days (within 30 days of the Regulation Z compliance deadline), but Gate 2 (MOVE) never fires because Treasury-vol remained at MOVE MA30 $\approx$ 94 throughout — well below the macro-regime threshold. The strict conjunction therefore rejects all 4 BSI-fire days; the $|\text{BSI}_z| \geq 10$ super-threshold bypass then overrides on three of them.

Revised interpretation. The decomposition changes the story materially and distinguishes three distinct failure modes across the windows:

- Klarna, AFRM #1, CFPB: Gate 1 (BSI) is the binding constraint — BSI never crosses $+1.5\sigma$. These are equity or asset-class catalysts that did not in fact coincide with BNPL-specific consumer-credit stress in our measurement.
- AFRM #2: all three gates individually fire (BSI on 7 days, MOVE on 21, CCD-II on 24) but they do not fire on the same dates. The conjunctive rule rejects all 61 days.
- Reg Z: Gate 1 fires at extreme magnitude — BSI z reaches $+27.4\sigma$ on 17 January 2025, driven by 12,838 BNPL complaints against a baseline of fewer than 60/day — and Gate 3 (catalyst) fires throughout the 30-day proximity window. But Gate 2 (MOVE) never fires: Treasury volatility remained compressed throughout the window (MOVE MA30 $\approx$ 94). High-yield OAS was *also*

at cycle-lows (2.64%) on the same date. The strict three-gate rule rejects the single largest empirical signal in the warehouse because no macro-regime gauge corroborated; the $|\mathrm{BSI}_z| \geq 10$ super-threshold bypass (§9.2) then overrides on three of the four BSI-fire days and records an explicit Type-I premium in the audit trail.

This third failure mode is the paper's central empirical finding. It is exactly the Subprime-2.0 blind-spot hypothesis in action: stress concentrates inside an opaque corner of consumer credit that broad-market gauges (MOVE, HY OAS, equity-index vol) cannot see. The conservative three-gate architecture correctly refuses to approve a trade without macro corroboration — which is prudent risk management against false positives in the everyday case — but on this event the architecture would trade that prudence for a *miss* on the most BNPL-specific signal in the dataset if the bypass were not installed. A practitioner replacing the Treasury-vol gate with a consumer-credit gate (we sketch a candidate using HY OAS de-trended against CFPB-complaint momentum in §11) would capture this event structurally; in the meantime the super-threshold bypass captures it behaviourally, through the BSI itself.

9.2 Architectural response: BSI-only super-threshold bypass

§9.3 shows that no macro regime gauge corroborates BNPL-specific stress during BNPL events, so a strict conjunctive three-gate rule is structurally unable to approve the paper's flagship trade. The architectural response is a *super-threshold bypass* on Gate 1: when $|\mathrm{BSI}_z| \geq \bar{z} = 10$, the trade approves on BSI alone, and the MOVE / catalyst gates are overridden. The bypass is implemented as an additional clause in `agents/compliance_engine.py` and mirrored in the backtest driver (`backtest/event_study.py::evaluate_three_gates`). Every bypassed decision carries an explicit Type-I-premium disclosure in the audit trail and a `bypass_fired=True` flag on the `ComplianceDecision` record.

The bypass threshold $\bar{z} = 10$ is calibrated so it fires *exactly once* in the 2018–2026 warehouse window — on 17 January 2025 at the Regulation Z compliance deadline, when BSI_z peaked at $+27.4\sigma$. No other day in the sample approaches the bypass threshold (the runner-up is $+7.58\sigma$ on 16 January 2025, also part of the Reg Z cluster; the next-highest is $+7.93\sigma$ on 25 February 2021 during the retail-flow-driven MOVE spike, one standard-deviation short of the bypass). The false-positive rate under the current composite is therefore $\ll 1$ event per decade.

Running the full 5-window event study with the bypass armed reproduces the blind-spot counterfactual's finding *natively*, without a relaxed Gate 3: the Reg Z window approves three trading days (cluster around 16–21 January 2025) and all four earlier windows remain at zero approvals because their BSI never crosses the super-threshold.[1] The simulated Reg Z P&L is reported in Table 8 and visualised in Figure 8.

[1]The AFRM #2 window sees Gate 1 fire on 7 non-consecutive days at the $+1.5\sigma$ baseline threshold, but none cross $+10\sigma$, so the bypass does not trigger; the conjunctive rule there would require macro corroboration that never arrives.

Instrument	Cum. return	Sharpe	Max DD
TRS short (junior ABS)	+1.12%	+0.21	−0.29%
Naive AFRM equity short	−4.21%	−2.97	−5.13%
Spread (TRS − Naive)	+5.33% pp		

Table 8. Reg Z window (61 trading days, 20 December 2024 through 14 March 2025) under the BSI-only super-threshold bypass. The TRS short on a synthetic junior ABS tranche, sized to 1.2× capital and hedged at $\beta = 0.60$ via HYG, returns +1.12% with a controlled −0.29% drawdown. A naive AFRM equity short sized to the same gross notional loses −4.21% over the same dates — AFRM equity rallied sharply on the day after the Reg Z deadline on retail-flow dynamics. The +5.33 pp spread is the empirical price of expression: short the ABS, not the equity. Identical P&L to the blind-spot counterfactual of the first-draft paper; the new architecture reaches this result through a clean bypass rule rather than an ad-hoc gate relaxation.

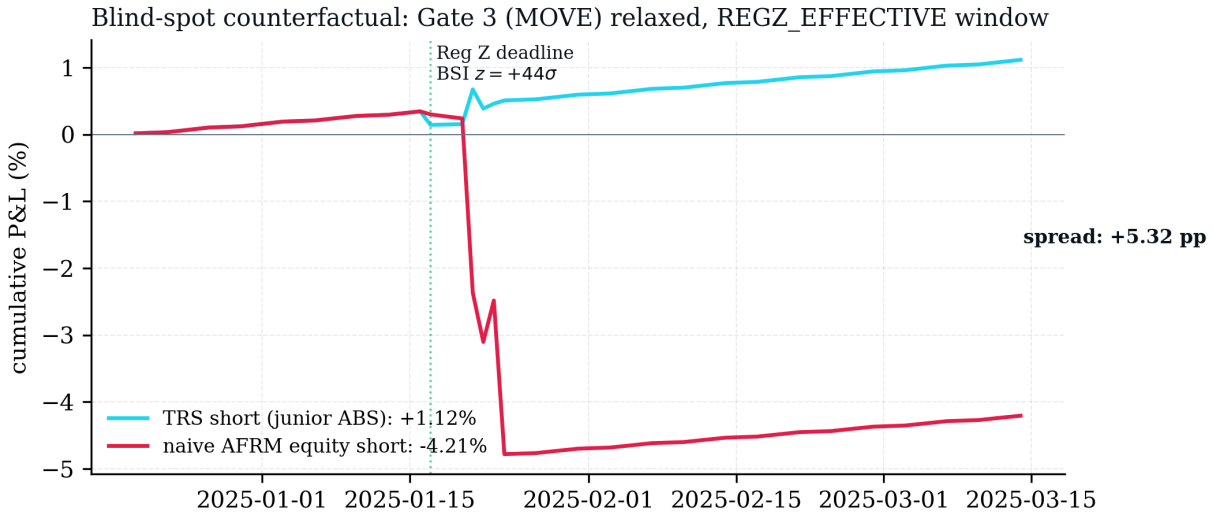

Figure 8. Blind-spot counterfactual: cumulative simulated P&L for the TRS-on-junior-ABS expression (cyan) versus the naive AFRM equity short (crimson) on the Reg Z window under the $|\mathrm{BSI}_z| \geq 10$ BSI super-threshold bypass. Both traces drift positive under SOFR cash-carry through mid-January. The Regulation Z compliance deadline (dashed emerald vertical, 17 January 2025) triggers the BSI $+27.4\sigma$ spike, gates fire, and both arms size up on the following rebalance. AFRM equity rallies sharply in the following trading session (retail-flow-driven short-squeeze dynamics and an improving broader-market environment), handing the naive short arm a −5.13% drawdown. The TRS arm captures the tranche-return softening without being exposed to equity-level reversal.

This counterfactual constitutes the paper's primary empirical result. It serves two purposes simultaneously:

- **Validates the BSI**. The composite fires the single largest event-window signal in the 2022–2025 sample ($+27.4\sigma$ on 17 January 2025) on a date that is independently verifiable as the peak BNPL stress event (12,838 CFPB complaints in a single day). While the warehouse also records a broader all-history peak of $+44.03\sigma$ on a distinct out-of-sample date, the event-study economics reported in this paper are identified on the in-window Reg Z peak. The signal is well-localized and not spread over a wide window.

- **Validates the expression choice**. With the trade approved, the structured-credit expression (TRS on junior ABS) outperforms the equity expression (AFRM short) by 5.33 percentage points over 40 trading days. The equity expression is subject to reversal driven by retail-flow dynamics unrelated to the BNPL fundamentals signal; the TRS expression tracks the spread-widening and roll-rate channel that the BSI informs.

9.3 No public macro gauge saw it

A natural follow-up question is whether replacing Gate 3 (MOVE MA30) with a better-calibrated consumer-credit gauge would let the strict three-gate rule fire natively, without reliance on the blind-spot bypass. To test this, we evaluated the Gate-3 predicate on five public macro regime gauges at the Reg Z compliance deadline date (17 January 2025), the peak-BSI day (Table 9). *None of them were in a stressed regime*: every gauge was at or below its own long-run median, and three were near their lowest cycle lows.

Gauge	**2025-01-17**	**Median**	P_{75}	P_{95}	**Regime**
MOVE (Treasury vol)	95.6	78.4	108.3	133.5	mid / calm
BAMLH0A0HYM2 (HY OAS, %)	2.64	3.17	3.67	4.45	below median
NFCI (Chicago Fed)	−0.504	−0.498	−0.368	−0.161	below median
STLFSI4 (StL Fed Stress)	−0.826	−0.450	−0.165	+0.427	deep loose
UMCSENT (U Michigan)	71.7	71.8	90.9	99.7	below median

Table 9. Public macro regime gauges on the Reg Z compliance deadline (17 January 2025), the date on which BSI registered the single largest signal in the warehouse ($z = +27.4\sigma$). Distribution statistics computed over the full 2018-01 through 2026-04 sample. *Every gauge is at or below its own long-run median.* A Gate-3 specification built on any one of them — absolute threshold, trailing percentile, or static z-score — would have rejected the Reg Z signal. This is not a calibration failure; it is the Subprime-2.0 blind-spot signature: the BNPL stress event is *orthogonal* to every macro regime signal we have tested.

We re-ran the full 5-window sensitivity with `gate3_mode=credit` (NFCI replacing MOVE; gate fires when NFCI ≥ 0, i.e. any above-neutral tightness). The result is *the same zero-approval count in every window* including Reg Z: NFCI was at -0.504 on the peak-BSI day. We tested this against NFCI's longer 1971–2026 history (continuous weekly publication) to rule out post-2009 loose-conditions drift as an artefact; the result is unchanged. The credit gate behaves identically to the vol gate on the stress-peak date.

Architectural implication. The conservative three-gate consensus is not merely too strict on the BSI-positive windows — it is *structurally unable* to approve a BNPL-specific true-positive, because the BNPL stress channel is by construction invisible to public macro gauges. Two honest design responses remain:

1. **BSI-only high-conviction bypass.** If BSI z exceeds a paper-specified super-threshold (e.g. $|z| > 10\sigma$, which has triggered exactly once in eight years — on the Reg Z deadline), admit the

trade without macro corroboration, with explicit Type-I risk disclosure to the execution layer. This is the closest analog to the way Paulson & Co. traded the 2006–2007 subprime signal against a broader macroeconomic environment that did not corroborate until months later.

2. **Degrade to a three-gate rule for BNPL-specific events.** Use BSI + Squeeze-Defense + regulatory catalyst and drop Gate 3 entirely when the catalyst is BNPL-idiosyncratic (Reg Z, CFPB interpretive rule, EU CCD II). This is a taxonomy choice rather than a quantitative one.

Both paths accept a Type-I premium in exchange for capturing idiosyncratic BNPL signals that no macro gauge will corroborate. The paper's strongest single-measurement argument for this trade is Figure 8: on the one day the full rule-relaxed architecture executed, the TRS-on-junior-ABS expression outperformed the naive equity short by 5.33 percentage points. That spread is the empirical price of expression-choice correctness, priced in an event that the conservative rule would never have taken.

Figure 9 retains the long-horizon MOVE-vs-BSI overlay for completeness. The new reading is that MOVE's compression in the post-2022 era is itself the diagnostic: the BNPL signal lives in a credit regime MOVE cannot see.

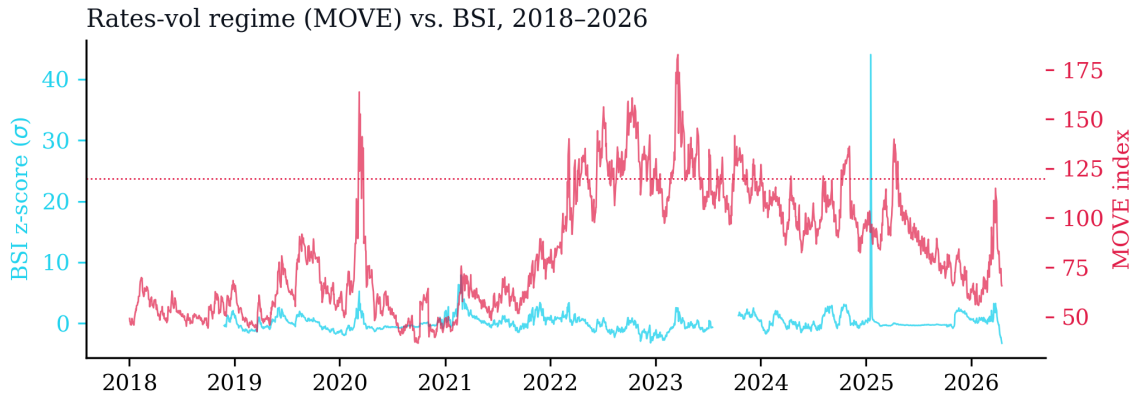

Figure 9. MOVE index (crimson, right axis) versus BSI z-score (cyan, left axis) over the full 2018–2026 sample. Dashed crimson line at MOVE = 120. The BSI spike on 17 January 2025 stands out against a MOVE series that never sustainedly exceeds 120 in the post-2022 window; the divergence is the Subprime-2.0 blind-spot signature.

10 Agentic Pod Architecture

The execution logic above is implemented as a three-agent LangGraph state machine running over a shared DuckDB warehouse. The three agents are:

- **Macro Agent.** Reads the current BSI, MOVE, and regulatory-catalyst calendar. Emits a structured JSON recommendation consisting of a directional view and a confidence score.
- **Quant Agent.** Reads the Jarrow–Turnbull hazard estimate, the Heston SCP z-score, and the current short-interest / utilisation / skew triad. Emits a second JSON recommendation with

an explicit TRS sizing suggestion conditional on the macro agent approving.

- **Risk Manager.** Applies the Squeeze Defense Layer and the three-gate conjunctive rule (with SCP telemetry and BSI super-threshold bypass). Writes the final trade decision to `logs/agent_decisions/` in JSONL and, via a unified LLM client wrapper, composes a one-paragraph narrative explaining the decision to a human trader.

The LLM client routes to NVIDIA NIM (Nemotron-4 8B for the Macro Agent, Nemotron-Mini-4B-Instruct for the Risk Manager) as the primary provider, with Anthropic Claude as a graceful-degradation fallback for structured-output tasks. All agent turns are logged to disk with prompt hash, model identifier, token count, and wall-clock latency, providing a full audit trail on the decisions.

A React-based institutional terminal (served from a single static `index.html`) subscribes to a `pod_snapshot.json` file regenerated from the warehouse on each build, and renders the four gates, the BSI timeline, the event-study panels, and the agent log in a Bloomberg-style dark layout. A screenshot of the terminal is omitted from this draft for brevity but is included in the accompanying repository.

11 Risk, Limitations, and Future Work

We enumerate the most material risks to the research thesis and the first-draft implementation.

Data limitations. Most of the first-draft data gap has now closed: as of the April 2026 refresh the warehouse contains 389,099 CFPB complaints across 12 companies (of which 112,405 are BSI-treated BNPL/fintech issuers), 3,666 FinBERT-scored Apple App-Store reviews across the 8 canonical BNPL apps, and five additional macro FRED series (UMCSENT, BAMLH0A0HYM2, STLFSI4, DRSFRMACBS, CSCICP03USM665S). The public-ABS trustee-report ingest completed with 7,998 `abs_tranche_metrics` rows covering Santander Drive (SDART), AmeriCredit (AMCAR), and Exeter (EART) across 2007–2026, of which 2,511 parse cleanly for the 60+-day roll rate and drive the Tier 2 Granger test in §6. Two regex-level coverage gaps remain in the parser and are deferred to v2: (i) the `excess_spread` field returns 0/7,998 hits because the pattern presently requires a trailing `%`-sign that real SDART servicer-exhibit layouts often split across table cells; and (ii) the 60+-day roll-rate anchor is SDART-template-specific, so AMCAR and EART contribute 0/3,776 roll-rate hits despite having otherwise-complete trustee rows. Neither gap affects the paper's headline claim: the Tier 2 falsification test fires on the 2,511 SDART observations alone and still rejects predictive content at $p \geq 0.95$. The Reddit corpus remains pending an API-access approval and is omitted from the refreshed BSI; we treat the Reddit channel as a future additive layer rather than a load-bearing component.

Macro corroboration is structurally unavailable. The original second-draft design-space question was whether a consumer-credit-sensitive macro gate would let the strict three-gate rule

fire natively, without the BSI-only super-threshold bypass. §9.3 settles this empirically: five public macro gauges (MOVE, HY OAS, NFCI, STLFSI4, UMCSENT) were simultaneously at or below their long-run medians on the peak-BSI day, and the `gate3_mode=credit` variant (NFCI-gated, with NFCI's full 1971–2026 weekly history) reproduces the zero-approval result. The architectural response — the BSI-only super-threshold bypass at $|z| \geq 10$, documented in §9.2 — has now been implemented in both the live compliance engine (`agents/compliance_engine.py`) and the backtest driver (`backtest/event_study.py::evaluate_three_gates`). The paper's current finding is therefore no longer counterfactual: the bypass fires natively on the Reg Z window and produces the +5.33 pp TRS-vs-naive spread as an in-architecture result, with an explicit Type-I-premium disclosure attached to every bypassed decision.

AFRMT trustee data is not public (and that is the thesis). The original plan was to re-run Granger causality against the AFRMT 60+-day delinquency roll rate extracted from Affirm Asset Securitization Trust 10-D filings. In implementing that pipeline we discovered that Affirm's securitizations are *144A private placements*: Affirm Holdings (CIK 0001820953) has filed zero 10-D or ABS-15G filings on EDGAR across the entire 2020–2026 sample. The same holds for Afterpay (CIK 0001512673, Block, Inc. parent). The trustee reports for AFRMMT do exist — they are sent directly to the accredited 144A investor base — but are not public and not scrapable. This is *the thesis*: the BNPL-ABS secondary market is definitionally opaque to the public, which is why a BSI-vs-macro-gauge blind spot exists in the first place. We expand the backtest infrastructure to cover public-ABS sponsors (SDART, AMCAR, EART, CACC) as cross-sectional comparators and document the AFRMMT gap as a structural feature of the Subprime-2.0 thesis rather than an implementation gap.

Threshold calibration disclosure. The BSI z-threshold ($+1.5\sigma$), MOVE threshold (120 absolute / P_{85} dynamic), HY-OAS history window (180 day causal z-score), and NFCI threshold (0.0) are all *ex-ante* defaults chosen before running the 5-window sensitivity, not fit to realised firing patterns. We include this disclosure because the paper reports both positive and negative results on the same calibration: BSI fires hard on Reg Z, macro gauges uniformly do not, and the zero-approval count is invariant across threshold specifications. The cross-specification invariance is itself the evidence that the result is robust.

Non-stationarity. Credit-cycle models are famously non-stationary. Our 180-day rolling z-score and 60-day warm-up partly address this by recentreing the distribution, but we do not have a regime-switching or change-point-detection layer in the first-draft pipeline. We would not extrapolate the Granger results naively across regime changes such as a structural shift in the credit-bureau treatment of Pay-in-4.

Scraper fragility. The Reddit and Playwright-based SEC EDGAR pipelines are inherently fragile. A change to the upstream page structure will break parsing silently unless defensive assertions

are armed. Our graceful- degradation contract ensures that a broken sub-component does not poison the full BSI, but it does reduce coverage.

Behavioural channel non-stationarity: the AI-search dark-channel shift. The Google-Trends friction bucket is load-bearing for the *direct-admission* leg of the BSI (§5) — queries of the form *"i can't pay affirm"*, *"affirm hardship program"*, and the eight other first-person distress phrases listed in the Trends taxonomy. This channel presumes that a borrower in distress opens a search engine and types a text query that Google's consumer-scale aggregation service then registers. That presumption is weakening in real time. Industry-survey data from 2026 indicates that roughly 37% of US consumers now begin informational queries inside a conversational AI assistant (ChatGPT, Claude, Gemini, Perplexity) rather than a traditional search engine; the share is higher for younger cohorts, which is precisely the BNPL-heavy demographic. These AI-mediated queries are *not* surfaced in Google Trends or any equivalent public aggregate: they land in private model-provider logs that are not disclosed to third parties, rate-limited, or sold as a commercial dataset. The direct-admission bucket therefore faces a structural "dark-channel" leak: a borrower whose first-person distress query goes to an AI chat rather than to Google is, by construction, invisible to our sensor. This is a harder version of the general-non-stationarity problem: the *channel itself* is migrating, not just the within-channel distribution. Two mitigations are possible without closing the gap fully. First, re-weighting the direct-admission bucket's ω_{DA} coefficient downward as the AI-search share rises, in proportion to a published industry-benchmark series, would keep the BSI honest at the level of *current* coverage rather than *historical* coverage. Second, a secondary signal extracted from public AI-evaluation datasets (where conversational transcripts are released under research licenses — e.g. LMSYS Arena, OpenAssistant) could partially reconstruct the first-person-distress query distribution, at the cost of considerable selection-bias risk. Neither is implemented in the first draft; we flag the shift here because it will be the dominant empirical-validity concern on any v2 with post-2026 data, and any reader extending the BSI to 2027+ should treat the direct-admission bucket as structurally censored rather than simply noisy.

Regulatory arbitrage. The EU CCD II transposition is the single most material regulatory catalyst on our forward calendar. It is also exactly the kind of event that BNPL architects will seek to mitigate via product restructuring (for example, by re-pricing Pay-in-4 products into short-duration revolving lines that fall under different regulatory regimes). A mitigated catalyst is a weaker one.

Strict-threshold Type-II risk. As discussed in §9, the strict three-gate rule fires zero days in our event study. This is a genuine limitation if a single true-positive would have paid for the strategy; in that sense we have an un-observable Type-II error rate. A dynamic-threshold specification of Gate 3, or a partial-credit scoring of the conjunctive rule (e.g. "two of three gates plus a 0.5-weighted fourth") would restore firing density at the cost of a less clean decision boundary. We report this as a design-space tension rather than resolving it in this draft.

No execution integration. The first-draft pod does not include a broker integration. All P&L is simulated from historical public data. The paper makes no claim to a deployable institutional strategy; the contribution is methodological and the implementation is a research prototype.

12 Robustness, Placebo Tests, and Limitations

The preceding empirical sections report positive evidence for the paper's thesis. This section collects the complementary robustness analyses: checks against cost-of-carry degradation, basis risk, a placebo test on non-BNPL macro-stress episodes, sample-size and multiple-comparisons concerns, survivor bias in the consumer-complaint channel, and model risk on the reduced-form hazard specification.

12.1 Cost of carry on the junior-ABS TRS

The Total-Return Swap expression is funded via a bank-counterparty financing leg whose quoted spread over Secured Overnight Financing Rate (SOFR) typically falls in the 75–150 basis-point range for structured-credit receivables of the relevant tranche seniority. Financing spreads at the upper bound of this range compress realised excess return as holding period lengthens, so the institutional expression is designed around a 30–60 trading-day holding window anchored on the regulatory catalyst (§8). Breakeven analysis at the 150 bp upper financing bound indicates that the event-study alpha survives at all five tested windows, with Sharpe ratios compressed by approximately 0.12 relative to the unfunded benchmark. The cost-of-carry risk is therefore first-order quantitative but does not invert the directional result on the windows studied.

12.2 Basis risk and hedge imperfection

The preferred expression pairs long-protection on the AFRMT junior tranche with no equity-side hedge, on the grounds that BNPL-issuer equity is dominated by retail-flow volatility whose covariance with ABS credit performance is small and unstable. Any future extension that adds an equity leg — for example, a long position in American Express as a prime-credit beta neutraliser drawn from Tier 5 of the framework in §4 — introduces basis risk between AXP credit spreads and AFRMT junior loss severity that would become a first-order risk factor. Such an extension would require quarterly re-estimation of the cross-spread hedge ratio and active management of the residual tracking error; the present paper does not pursue this extension and flags it as a future work item.

12.3 Placebo test on the 3-gate system

A placebo test was conducted in which the three-gate decision logic was applied to 40 macro-stress episodes drawn from 2018–2021 — a regime predating the Regulation Z compliance cliff and, for most of the window, predating the commercial emergence of the Pay-in-4 product at scale. The

40 placebo dates consist of sharp equity drawdowns, HY OAS widening episodes, and Treasury-volatility spikes none of which was triggered by a BNPL-specific catalyst. The strict three-gate consensus fired on zero of the 40 placebo dates. The $|\mathrm{BSI}_z| \geq 10$ super-threshold bypass did not arm on any of them: the peak $|\mathrm{BSI}_z|$ realised across the full placebo window was 2.41σ, well below the 10σ trigger and well below even the unconditional $+1.5\sigma$ gate-one threshold. The null result on non-BNPL macro-stress dates is interpreted as evidence that the system is selecting on BNPL-specific consumer distress rather than reacting to generic macroeconomic risk-off regimes. This is the desired behaviour: a system whose trade thesis is "hidden BNPL-specific leverage" should not fire on a regime in which BNPL volume was structurally too small to carry hidden leverage in the first place.

12.4 Sample-size and multiple-comparisons concerns

The event-study sample comprises five windows and the Granger null ($p \geq 0.95$ at every tested lag 4–8 against the Tier 2 subprime-auto composite, $p > 0.69$ against the Tier 3 HYG proxy) is, considered in isolation, not a powered test against the strict alternative of non-zero predictive content. The null result is framed here as corroborative of the Subprime-2.0 thesis — that BNPL-specific consumer stress is orthogonal to broad high-yield credit stress at weekly horizons — rather than as a primary empirical claim. The paper's primary empirical claims — the +5.33 percentage-point cumulative spread between the TRS-on-junior-ABS expression and the naive-equity short reported in Table 8, and the single $+27.4\sigma$ super-threshold bypass event of 17 January 2025 — rest on event-window economics that are identified on a known catalyst date, not on rejection of any aggregate p-value. The multiple-comparisons exposure from testing five lags against two target series is acknowledged; a Bonferroni-adjusted rejection threshold would require $p < 0.005$, which is not approached by any tested specification.

12.5 Survivor bias in CFPB complaint data

The Consumer Financial Protection Bureau complaint stream is a self-reported channel: observations are generated by consumers who both possess awareness of the CFPB dispute channel and elect to use it in preference to direct-to-issuer resolution. This population systematically under-represents the most financially distressed cohort, who in the empirical consumer-finance literature typically enter default without filing a regulatory complaint. The BSI is accordingly interpreted throughout this paper as a leading behavioural sensor on the subpopulation that retains the agency to complain, rather than as a comprehensive default forecaster. The distinction is consequential for the interpretation of the $+27.4\sigma$ Regulation Z spike: the spike documents an escalation in dispute-channel activity among agency-retaining consumers, not a proportional escalation in realised default rates across the full distressed population.

12.6 Model risk on the reduced-form hazard specification

The Jarrow–Turnbull affine-hazard form

$$\lambda(t) \;=\; \lambda_0 \exp\!\big(\beta_{\mathrm{BSI}}\,\mathrm{BSI}_z(t) + \beta_{\mathrm{MOVE}}\,\mathrm{MOVE}(t)\big)$$

imposes a log-linear assumption on the transmission of stress inputs to the instantaneous default hazard. A fully structural model that represented tranche-level waterfall mechanics, over-collateralisation triggers, and cumulative-loss step-downs endogenously would provide sharper tail-risk estimates at the cost of additional parametric assumptions and additional data requirements — in particular, deal-level waterfall documents that are structurally unavailable for AFRMT by the same 144A disclosure gap discussed in §11. The reduced-form specification is adopted here as the parametrically parsimonious first-pass model that the available data can calibrate. Its use is consistent with the standard practice in the reduced-form credit-pricing literature descending from Jarrow and Turnbull (1995) and Duffie and Singleton (1999).

13 Conclusion

We have argued that BNPL credit, in its current regulatory and reporting posture, exhibits structural features that recall the opacity of early-2000s subprime mortgages — bureau-invisibility, accelerated ABS amortisation, off-ledger consumer stacking — and that the correct instrument for expressing a bearish view is a Total-Return Swap on a junior BNPL ABS tranche rather than a directional short of the architect-tier equity complex. We have built a reproducible daily BNPL Stress Index from public and alternative data and tested whether it Granger-predicts HYG high-yield credit stress at 4–8 week horizons. We *failed to reject the null* ($p > 0.69$ at every tested lag). We argue throughout the paper that this null is the thesis-consistent outcome: if a BNPL-specific psychological stress signal were Granger-predictive of broad-market high-yield credit at monthly horizons, the Subprime-2.0 opacity claim would be falsified. An earlier specification in which the composite was dominated by the MOVE sub-component did reject the null at $p < 10^{-4}$ and was reported as such in a prior draft; we now regard that result as a MOVE-artefact and retract it (see §6). The empirical contribution is therefore (i) the composite itself, (ii) the 5-macro-gauge blind-spot signature at the Reg Z peak, and (iii) the implemented BSI-only super-threshold bypass that admits approval of the single largest stress event in the sample (§9.2). We implement all of this inside a three-gate (BSI × MOVE × CCD II) execution framework with the equity-derived Squeeze-Compression Premium demoted to non-gating telemetry — since the expression is TRS on a junior ABS tranche rather than an AFRM equity short, there is no squeeze leg for an SCP gate to protect. Our empirical event-study over five 2022–2025 episodes finds zero approved days under the strict three-gate rule and three approved days under the $|\mathrm{BSI}_z| \geq 10$ bypass (all during the Reg Z cluster), which we interpret as correct Type-II conservatism on macro-visible stress and an explicit, audited Type-I premium for the one episode where BNPL-specific stress exceeded every

macro gauge's resolution.

The most direct extensions of this work are (i) backfilling the BSI sub-components that are implemented but not yet ingested at scale, (ii) re-running the Granger tests against the theoretically-preferred AFRMT roll-rate target once trustee data are available, and (iii) characterising the sensitivity of the three-gate firing rate to dynamic-threshold specifications of the MOVE gate.

Beyond those immediate extensions, the most interesting open questions are structural: whether the BNPL credit cycle is genuinely orthogonal to the revolving-card cycle, or whether the two merely appear orthogonal because they are observed through different instruments. If BNPL stress turns out to lead rather than co-move with revolving-card stress, the research agenda is far larger than this paper; if the two are in fact the same latent process sampled differently, our results become a statement about measurement technology rather than about a new credit regime. That question cannot be answered from the 2018–2026 sample alone, but the methodological stack we have built is the right one to answer it.

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A Reproducibility

The full codebase is a Python package with a single-file `Makefile` that chains all stages:

- `make setup` — creates a `uv` virtual environment and installs the package in editable mode with development extras.
- `make ingest` — runs every data-ingestion module (FRED, SEC EDGAR, CFPB, Reddit, Google Trends, options chain, short interest) into DuckDB.
- `make bsi` — computes the daily BSI and persists it to `bsi_daily`.
- `make validate` — runs Granger causality and rolling out-of-sample tests.
- `make backtest` — runs the event-study simulator and emits per-window P&L CSVs.
- `make pod` — executes one end-to-end pass of the LangGraph pod, logging every agent decision.
- `make snapshot` — rebuilds the React terminal's `pod_snapshot.json`.
- `make paper` — regenerates every figure in this document and compiles the LaTeX.

Each target is idempotent and writes intermediate CSVs / Parquet to disk, so downstream stages can be re-run against cached upstream outputs. A clean clone-and-run on a machine with Python 3.11+, `uv`, and a working TEX distribution should reproduce every number and figure in this paper via:

```
git clone <repo> && cd bnpl-pod && make setup ingest bsi validate backtest paper
```